TASK - 2

ALGORITHMIC PROBLEM SOLVING 2

Ex 2.1 : Sort all the functions below in increasing order of asympotic growth. If some have the same asympotic growth, the be sure to indicate that

Note : log means base

I. $5n$ II. $4\log n$ III. $4 \log \log n$ IV. n^4 V. $n^{1/2}\log^4 n$
VI. $(\log n)^{5\log n}$ VII. $n^{\log n}$ VIII. 5^n IX. 4^{n^4} X. 4^{4n} XI. 5^{5n}
XII. $n^{n^{1/5}}$ XIV. $n^{n/4}$ XV. $(n/4)^{n/4}$

Objective :

Sort the given functions to increasing order of asympotic growth.

Key Rules :

- Ignore constants (4, 5 do not matter)
- log means log base 2
- Compare growth as $n \rightarrow$ infinity

Step-1 : Logarithmic Functions

$\log \log n$ grows slower than $\log n$

$4 \log \log n < 4 \log n$

Step-2 : Polynomial with log factor

$n^{(1/2)} \log^4 n$ grows faster than $\log n$ but slower than n.

$4 \log \log n < 4 \log n < n^{(1/2)} \log^4 n$

Step-3 : Linear and polynomial

$5n$ is linear

n^4 is polynomial

$\therefore < n^{(1/2)} \log^4 n < 5n < n^4$

Step-4 : Super polynomial

$(\log n)^{(5\log n)} = n^{(5\log\log n)}$

$\cdots < n^4 < (\log n)^{(5\log n)}$

Step-5 : $n^{(\log n)}$

$n^{(\log n)} = 2^{((\log n)^2)}$

$\cdots < (\log n)^{(5\log n)} < n^{(\log n)}$

Step-6 : Power Growth

$n^{(n^{(1/5)})}$ grows faster than $n^{(\log n)}$

$\cdots < n^{(\log n)} < n^{(n^{(1/5)})}$

Step-7 : Exponential

$5^n < 4^{(4n)} < 5^{(5n)}$

Step-8 : Super Exponential

$(n/4)^{(n/4)} < n^{(n/4)}$

Step-9 : Fastest

$4^{(n^4)}$

Final Sorted order :

$4\log\log n$

$< 4\log n$

$< n^{(1/2)}\log^4 n$

$< 5n$

$< n^4$

$< (\log n)^{(5\log n)}$

$< n^{(\log n)}$

$< n^{(n^{(1/5)})}$

$< 5^n$

$< 4^{(4n)}$

$< 5^{(5n)}$

$< (n/4)^{(n/4)}$

$< n^{(n/4)}$

$< 4^{(n^4)}$

Ex 2.2 order the functions according to their growth from slowest to fastest growing.

$6n^3, n\log_6 n, 4n, 8n^2, \log_2 n, n\log_2 n, \log_8 n, 64, 8^{2n}$

Objective : order the given functions from slowest growing to fastest growing.

Given Functions:

- $6n^3$
- $n\log_6 n$
- $4n$
- $8n^2$
- $\log_2 n$
- $n\log_2 n$
- $\log_8 n$
- 64
- 8^{2n}

Important Rules:

- Ignore constant multipliers (4, 6, 8, 64)
- All algorithms differ only by constant factors.
- compare growth as $n \to \infty$

Step-1 : constant function

64 is a constant

It does not depend on n, so it grows the slowest.

64

Step-2 : Logarithmic Functions

$\log_2 n$ and $\log_8 n$ differ only by a constant factor

so they grow at the same rate

$64 < \log_8 n < \log_2 n$

Step-3 : Linear Function

$4n$ is linear

Linear grows faster than logarithmic

$\cdots < 4n$

Step-4 : Linearithmic Functions

$n \log_6 n$ and $n \log_2 n$ grow faster than linear but slower than quadratic.

Both have same asympotic growth

$\cdots < n \log_6 n < n \log_2 n$

Step-5 : Polynomial Functions

$8n^2$ is quadratic

$6n^3$ is cubic

Higher power grows faster.

$\cdots < 8n^2 < 6n^3$

Step-6 : Exponential Function

$8^{2n} = 64^n$ is exponential

Exponential growth dominates all polynomial functions.

Fastest growth.

Final sorted order (slowest → Fastest):

64

$< \log_8 n$

$< \log_2 n$

$< 4n$

$< n \log_6 n$

$< n \log_2 n$

$< 8n^2$

$< 6n^3$

$< 8^{2n}$

VEL TECH - CSE	
EX NO.	2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
SIGNATURE DATE	12/2/26

RESULT : There fore the above task were sucessfully executed & verified